

PROJECT DOCUMENTATION

# AI Hospital Priority Management System

*Intelligent Healthcare Triage & Resource Optimization Platform*

|                |                    |
|----------------|--------------------|
| Version        | 1.0                |
| Date           | June 2026          |
| Classification | Internal Use       |
| Status         | Active Development |

# 1. Executive Summary

The AI Hospital Priority Management System is an intelligent healthcare platform designed to transform patient triage and treatment prioritization in modern hospital environments. By leveraging Artificial Intelligence (AI) and Machine Learning (ML) techniques, the system evaluates patient symptoms, medical history, and real-time clinical indicators to assign automated priority levels — ensuring the most critical patients receive immediate attention.

This system addresses a critical gap in hospital operations: the delay between patient arrival and appropriate care allocation. Through automation and intelligent decision support, it reduces waiting times, optimizes resource allocation, and empowers medical staff with data-driven insights.

## Key Benefits

| Benefit                 | Description                                                     |
|-------------------------|-----------------------------------------------------------------|
| Faster Emergency Care   | Automated triage reduces decision time for critical patients    |
| Improved Accuracy       | AI-driven analysis minimizes human error in severity assessment |
| Resource Optimization   | Smart scheduling and allocation reduces operational waste       |
| Better Patient Outcomes | Timely prioritization directly improves treatment results       |
| Staff Decision Support  | Doctors receive AI-generated insights and recommendations       |

## 2. Project Overview

### 2.1 Problem Statement

Hospital emergency departments frequently face overcrowding, delayed triage, and inconsistent prioritization of patients. Manual processes are time-consuming and prone to human error, especially under high-volume conditions. The lack of a real-time intelligent system results in:

- Delayed treatment for critical patients
- Inefficient use of medical staff and resources
- Increased waiting times and patient dissatisfaction
- Higher risk of adverse medical outcomes

### 2.2 Proposed Solution

The AI Hospital Priority Management System automates the triage process through a multi-layered AI/ML pipeline. The solution includes:

- A Patient Interface for symptom submission and registration
- An AI Analysis Engine that evaluates clinical indicators in real time
- A Doctor Panel for reviewing AI-generated recommendations
- An N8N Workflow Engine for process automation and integration
- Appointment and booking management modules

### 2.3 Project Scope

| Module             | Scope                                      | Status         |
|--------------------|--------------------------------------------|----------------|
| Patient Interface  | Registration, symptom input, bookings      | In Development |
| AI Analysis Engine | ML model, priority scoring, result display | In Development |
| Doctor Panel       | Profile management, patient queue, login   | In Development |
| Appointment System | Scheduling, recent bookings, history       | In Development |
| N8N Workflow       | Automation, integrations, notifications    | In Development |

### 3. System Architecture

#### 3.1 Component Overview

| Component            | Description                               | Technology            |
|----------------------|-------------------------------------------|-----------------------|
| Patient Interface    | Web-based portal for patient interaction  | React / Web UI        |
| AI Analysis Engine   | Core ML model for priority classification | Python / ML Framework |
| Doctor Panel         | Clinical dashboard for medical staff      | Web UI / REST API     |
| N8N Workflow Engine  | Process automation and system integration | N8N Platform          |
| Database Layer       | Patient records, appointments, history    | Relational / NoSQL DB |
| Notification Service | Alerts, reminders, and real-time updates  | Email / SMS / Push    |

#### 3.2 AI Analysis Pipeline

The AI engine follows a structured five-step pipeline to evaluate and prioritize patients:

- Step 1 — Data Ingestion: Patient symptoms, vitals, and medical history are collected via registration
- Step 2 — Pre-processing: Data is cleaned, normalized, and encoded for ML model input
- Step 3 — Model Inference: The trained ML model evaluates the data and generates a priority score
- Step 4 — Priority Assignment: Patients are classified into urgency tiers (Critical, High, Medium, Low)
- Step 5 — Result Delivery: The analyzed result is displayed to both patient and doctor panel

## 4. Module Descriptions & Screenshots

### 4.1 Patient Interface

The Patient Interface is the primary touchpoint for individuals entering the hospital system. It is designed to be intuitive, accessible, and responsive across devices.

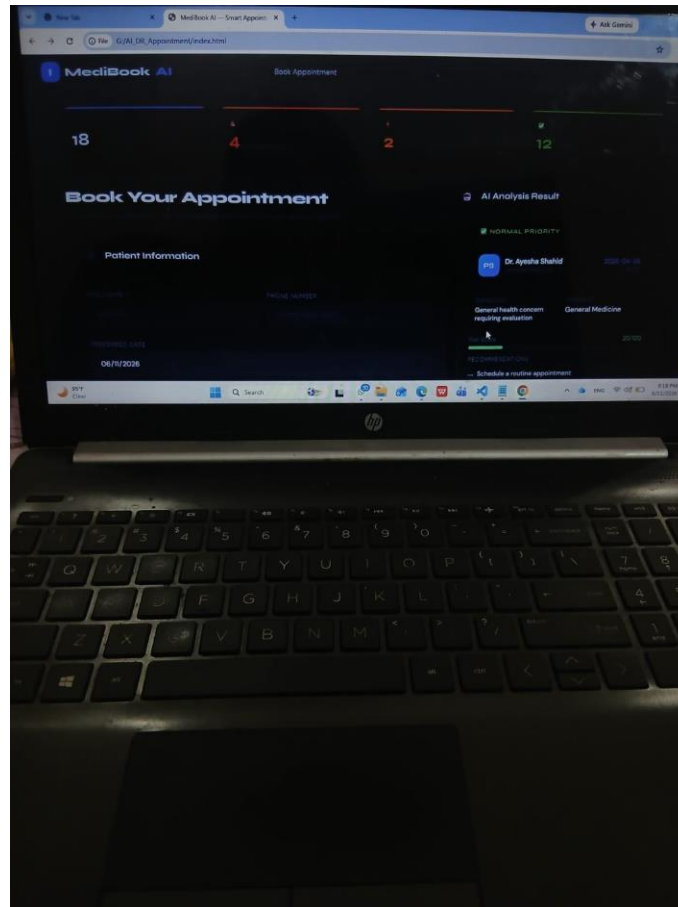
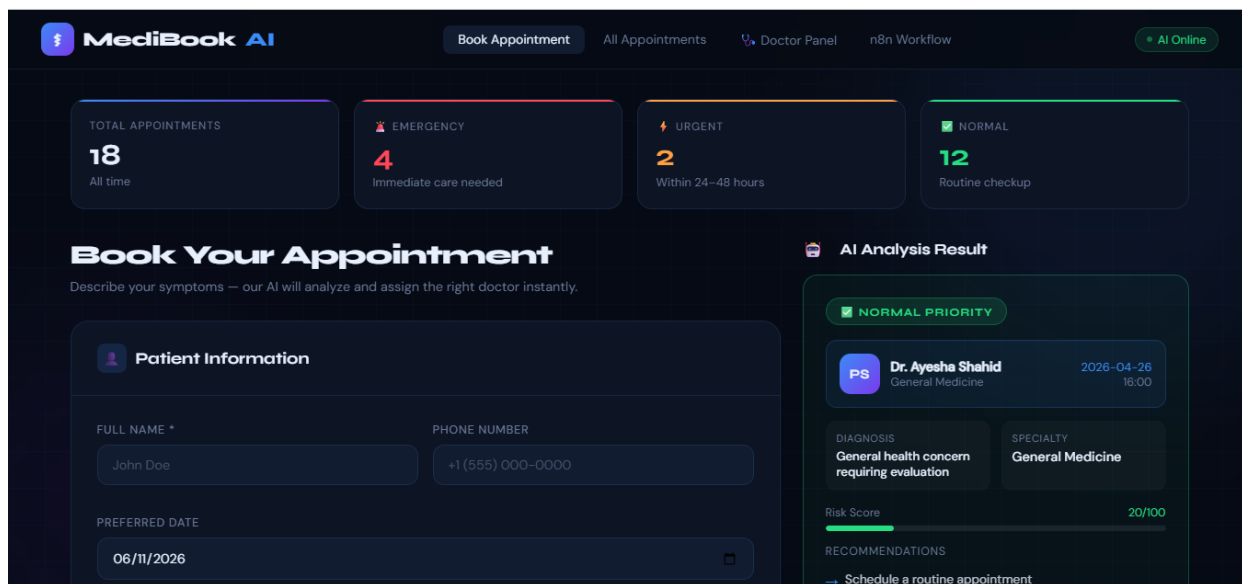


Figure 4.1a — Patient Interface (Mobile View)

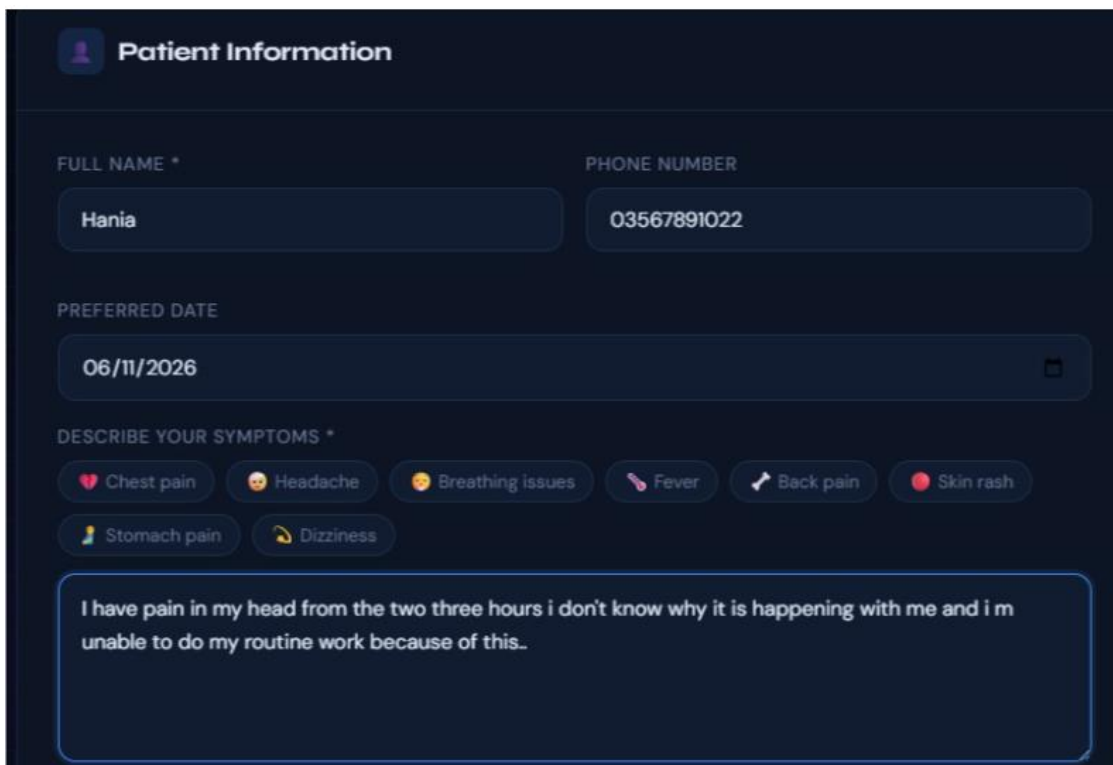


The screenshot shows the 'MediBook AI' patient interface. At the top, there's a navigation bar with 'Book Appointment', 'All Appointments', 'Doctor Panel', and 'n8n Workflow'. A status indicator 'AI Online' is in the top right. Below the navigation bar, there are four summary cards: 'TOTAL APPOINTMENTS: 18 All time', 'EMERGENCY: 4 Immediate care needed', 'URGENT: 2 Within 24-48 hours', and 'NORMAL: 12 Routine checkup'. The main section is titled 'Book Your Appointment' with a subtext 'Describe your symptoms — our AI will analyze and assign the right doctor instantly.' It features a 'Patient Information' form with fields for 'FULL NAME \*' (John Doe), 'PHONE NUMBER' (+1 (555) 000-0000), and 'PREFERRED DATE' (06/11/2026). To the right, an 'AI Analysis Result' panel shows 'NORMAL PRIORITY', a doctor 'Dr. Ayesha Shahid' (General Medicine) for '2026-04-26 16:00', a diagnosis of 'General health concern requiring evaluation', a specialty of 'General Medicine', a 'Risk Score' of '20/100', and a recommendation to 'Schedule a routine appointment'.

Figure 4.1b — Patient Interface Dashboard

## 4.2 Patient Registration

The registration module collects essential demographic and medical information required for AI-driven triage. Key data points captured include full name, chief complaint, reported symptoms, medical history, allergies, current medications, and vital signs.



The screenshot shows the 'Patient Information' registration form. It includes fields for 'FULL NAME \*' (Hania), 'PHONE NUMBER' (03567891022), and 'PREFERRED DATE' (06/11/2026). Below these is a section 'DESCRIBE YOUR SYMPTOMS \*' with buttons for 'Chest pain', 'Headache', 'Breathing issues', 'Fever', 'Back pain', 'Skin rash', 'Stomach pain', and 'Dizziness'. A text area at the bottom contains the patient's description: 'I have pain in my head from the two three hours i don't know why it is happening with me and i m unable to do my routine work because of this..'

Figure 4.2 — Patient Registration Form

### 4.3 AI Analyzing Result

Following registration, the system presents the AI-generated priority classification to the patient. This screen displays the assigned priority level with a color-coded severity badge, estimated wait time based on current hospital load, and recommended next steps.

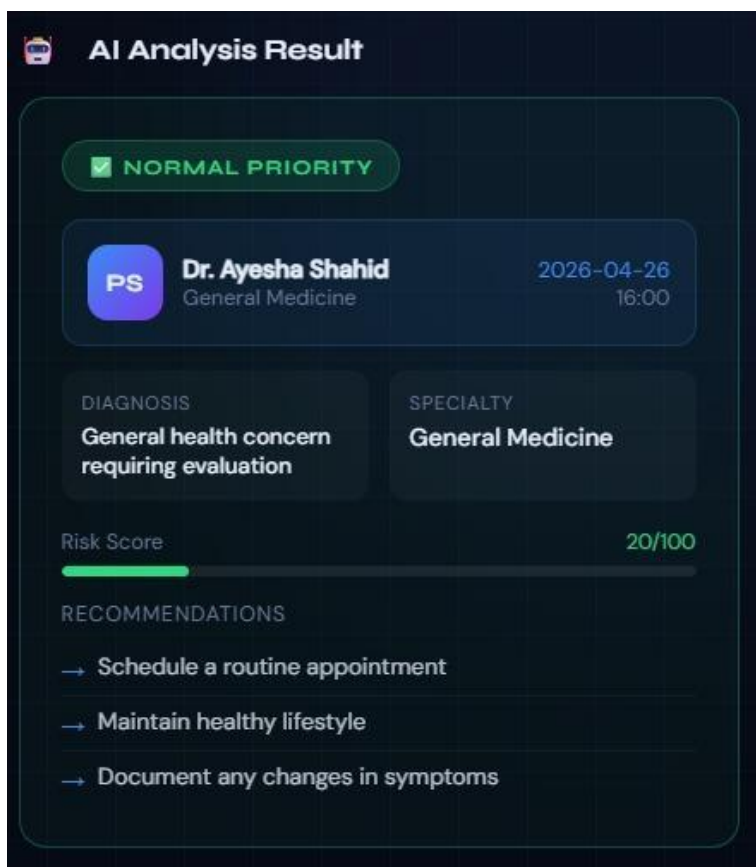


Figure 4.3 — AI Priority Analysis Result

### 4.4 All Appointments

This module provides patients with a complete view of their scheduled appointments, including upcoming visits and historical records. Features include appointment rescheduling, cancellation, and confirmation workflows.

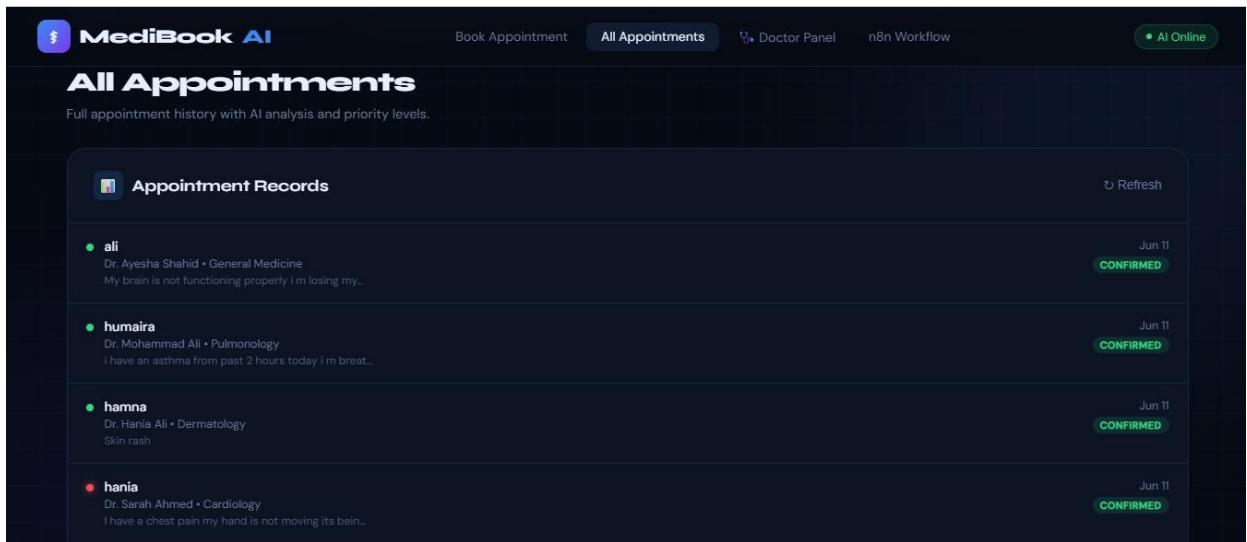


Figure 4.4 — All Appointments Screen

### 4.5 Recent Bookings

A summarized view of the patient's most recent booking activity, enabling quick access to upcoming appointments and associated medical instructions.



| Recent Bookings |                                                                                                         | Refresh   |
|-----------------|---------------------------------------------------------------------------------------------------------|-----------|
| ● ali           | Jun 11<br>Dr. Ayesha Shahid • General Medicine<br>My brain is not functioning properly i m losing my... | CONFIRMED |
| ● humaira       | Jun 11<br>Dr. Mohammad Ali • Pulmonology<br>I have an asthma from past 2 hours today i m breat...       | CONFIRMED |
| ● hamna         | Jun 11<br>Dr. Hania Ali • Dermatology<br>Skin rash                                                      | CONFIRMED |
| ● hania         | Jun 11<br>Dr. Sarah Ahmed • Cardiology<br>I have a chest pain my hand is not moving its bein...         | CONFIRMED |
| ● Hania         | Jun 11<br>Dr. Ayesha Shahid • General Medicine<br>I have pain in my head from the two three hours i ... | CONFIRMED |

Figure 4.5 — Recent Bookings View

### 4.6 Doctor Panel

The Doctor Panel serves as the clinical management dashboard, giving medical professionals full visibility into patient queues, AI recommendations, and appointment schedules.

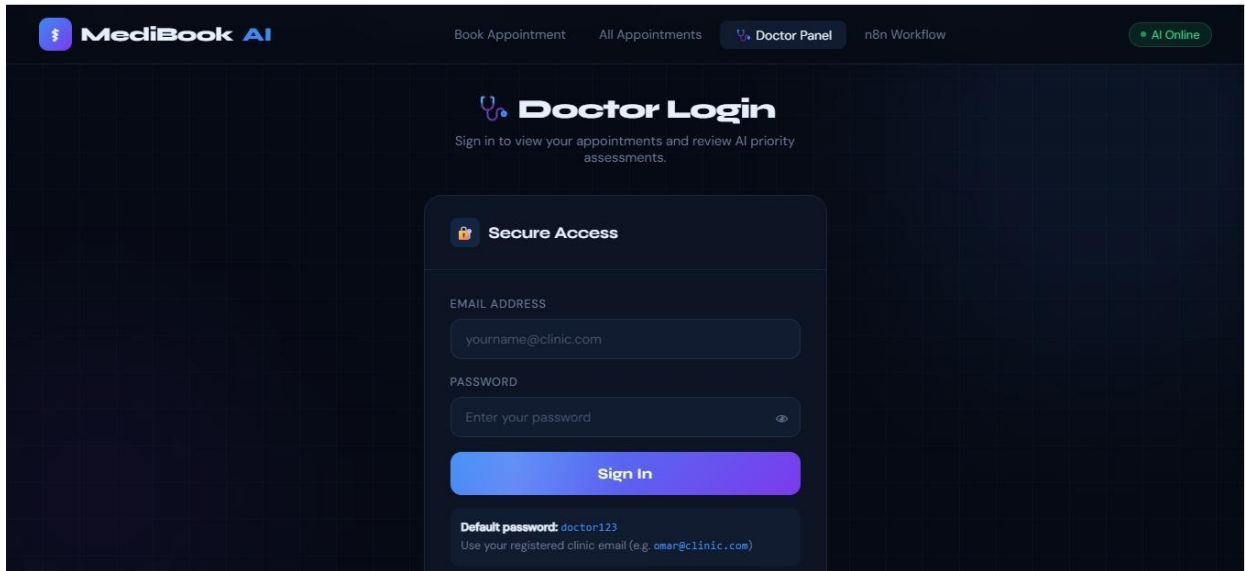


Figure 4.6 — Doctor Panel Dashboard

### 4.7 Doctor Login

Secure authentication module for medical staff. Supports role-based access control ensuring only authorized personnel access clinical data. Authentication is enforced via credential verification and optional multi-factor authentication.

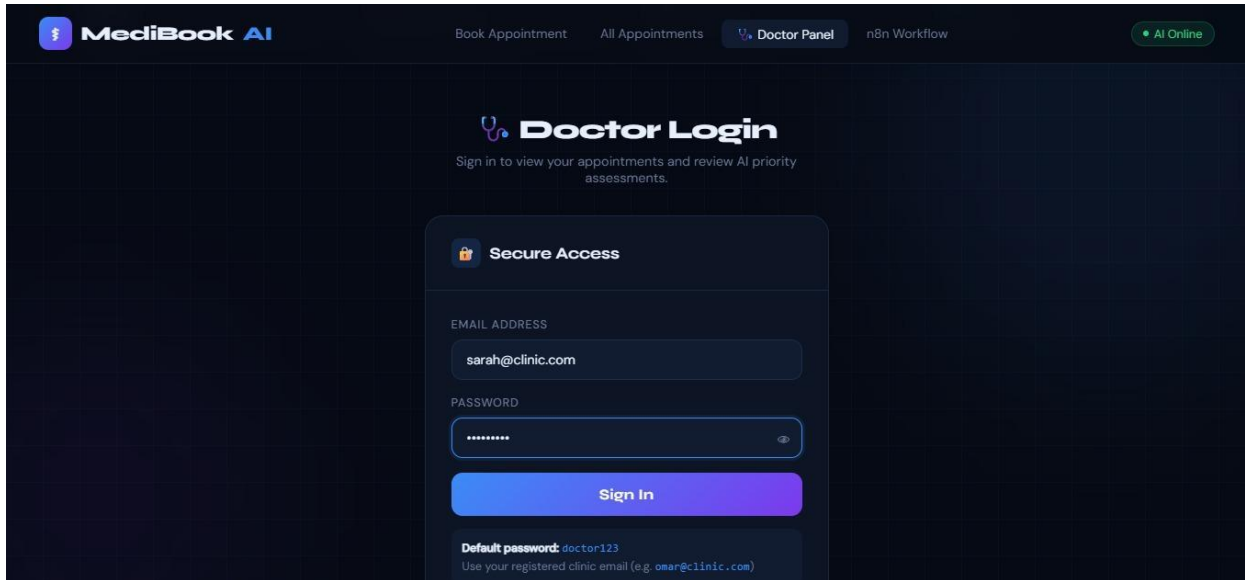


Figure 4.7 — Doctor Login Screen

### 4.8 Doctor Profile

Each registered physician maintains a professional profile containing their specialization, credentials, availability schedule, and assigned patient caseload. This data is used by the AI engine to intelligently route patients to the most appropriate available clinician.

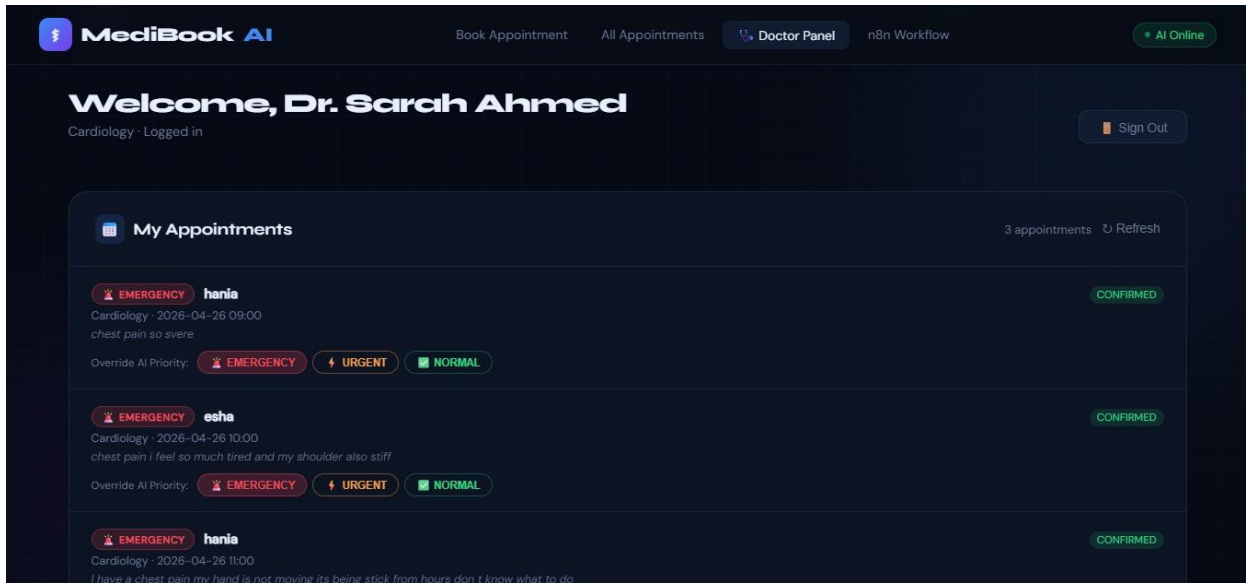


Figure 4.8 — Doctor Profile Management

## 4.9 N8N Workflow

The N8N workflow engine serves as the automation backbone of the system, orchestrating data flows and integrations between all modules. Key workflows include patient registration triggers, priority assignment notifications, appointment confirmations, and critical patient escalations.

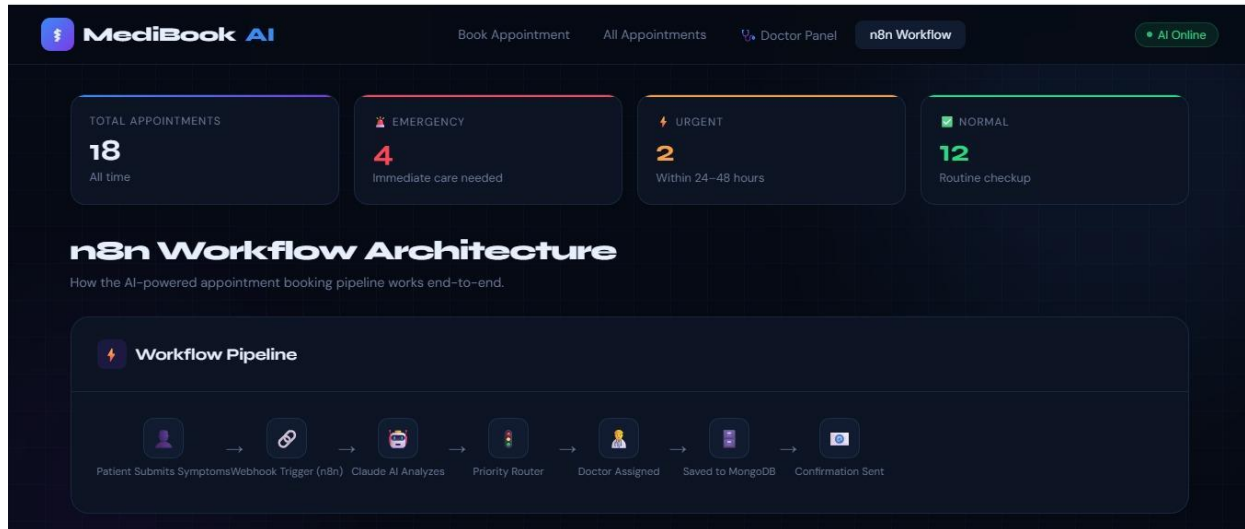


Figure 4.9 — N8N Automation Workflow

## 5. Setup & Installation Instructions

This section outlines the steps required to deploy and configure the AI Hospital Priority Management System in a development or production environment.

### 5.1 Prerequisites

| Requirement | Version              | Purpose                  |
|-------------|----------------------|--------------------------|
| Node.js     | v18 or higher        | Backend runtime          |
| Python      | 3.9 or higher        | AI/ML engine             |
| N8N         | Latest stable        | Workflow automation      |
| Database    | PostgreSQL / MongoDB | Data persistence         |
| Docker      | Optional             | Containerized deployment |
| Web Browser | Chrome / Firefox     | Frontend interface       |

### 5.2 Installation Steps

- Step 1: Clone the project repository from the source version control system
- Step 2: Install Node.js dependencies using `npm install` in the project root
- Step 3: Install Python dependencies using `pip install -r requirements.txt`
- Step 4: Configure environment variables by copying `.env.example` to `.env`
- Step 5: Initialize the database by running the provided migration scripts
- Step 6: Import the N8N workflow definitions from the `/workflows` directory
- Step 7: Start the application server using `npm start` or the designated startup command
- Step 8: Access the Patient Interface at the configured localhost port and verify all modules load correctly

### 5.3 Setup Reference Screenshot

The following screenshot illustrates the complete setup instructions as configured in the project environment:

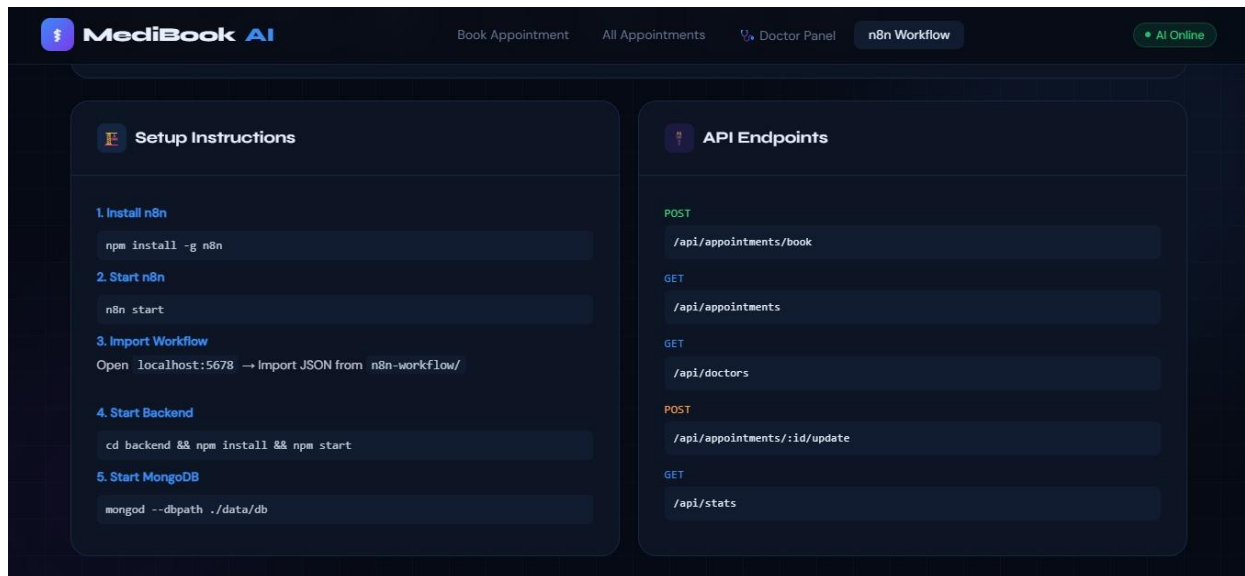


Figure 5.1 — System Setup Instructions

## 6. Functional Requirements

| ID    | Requirement               | Description                                                     | Priority |
|-------|---------------------------|-----------------------------------------------------------------|----------|
| FR-01 | Patient Registration      | System shall capture patient demographics and symptoms          | High     |
| FR-02 | AI Priority Scoring       | System shall classify patients into urgency tiers automatically | High     |
| FR-03 | Result Display            | System shall present AI results to patients in real time        | High     |
| FR-04 | Doctor Authentication     | System shall enforce secure login for medical staff             | High     |
| FR-05 | Patient Queue Management  | Doctor panel shall display prioritized patient list             | High     |
| FR-06 | Appointment Booking       | Patients shall be able to schedule and manage appointments      | Medium   |
| FR-07 | Recent Booking History    | System shall maintain and display recent booking records        | Medium   |
| FR-08 | N8N Workflow Triggers     | System shall trigger automated workflows on key events          | High     |
| FR-09 | Notification System       | System shall send alerts to patients and staff                  | Medium   |
| FR-10 | Doctor Profile Management | Doctors shall be able to maintain their professional profile    | Low      |

## 7. Non-Functional Requirements

| Category        | Requirement             | Target / Standard                                      |
|-----------------|-------------------------|--------------------------------------------------------|
| Performance     | AI response time        | Priority scoring completed within 3 seconds            |
| Scalability     | Concurrent users        | Support 500+ simultaneous users without degradation    |
| Availability    | System uptime           | 99.5% uptime SLA in production environment             |
| Security        | Data encryption         | All PHI encrypted at rest and in transit (AES-256)     |
| Security        | Access control          | Role-based access for patients, doctors, and admins    |
| Compliance      | Healthcare standards    | Adherent to applicable data privacy regulations        |
| Usability       | Interface accessibility | Accessible on desktop and mobile browsers              |
| Maintainability | Code documentation      | All modules documented with inline comments and README |



## 8. Glossary

| Term           | Definition                                                                          |
|----------------|-------------------------------------------------------------------------------------|
| AI             | Artificial Intelligence — simulation of human intelligence by computer systems      |
| ML             | Machine Learning — AI subset enabling systems to learn from data patterns           |
| Triage         | Medical process of prioritizing patients based on urgency of condition              |
| N8N            | Open-source workflow automation platform used for system integrations               |
| PHI            | Protected Health Information — sensitive patient data protected by regulation       |
| Priority Level | Urgency tier assigned to a patient (Critical / High / Medium / Low)                 |
| REST API       | Representational State Transfer — architectural style for web service communication |
| JWT            | JSON Web Token — secure method for transmitting authentication information          |
| EMR            | Electronic Medical Record — digital version of a patient's medical chart            |